21 Electrochemistry

21.1

To find the conventional cell reaction we use 'right – left', having made sure that there are the *same* number of electrons in the RHS and LHS half-cell reactions. How you choose to achieve this is up to you. For example, in (2) the solution given below chooses RHS – 2 × LHS, but you could just as well choose $\frac{1}{2}$ × RHS – LHS to give

$$\tfrac{1}{2}\mathrm{Fe^{2+}(aq)} + \tfrac{1}{2}\mathrm{H_2(g)} \longrightarrow \mathrm{H^+(aq)} + \tfrac{1}{2}\mathrm{Fe(m)}.$$

This latter choice means that only *one* electron is involved i.e. $n = 1$. The stoichiometric coefficients are different so the Nernst equation looks rather different to the one given below in Eq. 21.II:

$$E = E^\circ(\mathrm{Fe^{2+}/Fe}) - \frac{RT}{F} \ln \frac{([\mathrm{H^+}]/c^\circ)}{(p_{\mathrm{H_2}}/p^\circ)^{\frac{1}{2}}\,([\mathrm{Fe^{2+}}]/c^\circ)^{\frac{1}{2}}} \qquad (21.\mathrm{I})$$

However, careful comparison of Eq. 21.I and Eq. 21.II will show that they are identical (take the factor of $\frac{1}{2}$ from Eq. 21.II inside the log, where it becomes are square root).

1

$$\mathrm{Pt(m)|H_2(g)|H^+(aq)||Cu^+(aq)|Cu(m)}$$

$$\text{RHS} - \text{LHS} \quad \mathrm{Cu^+(aq)} + \tfrac{1}{2}\mathrm{H_2(g)} \longrightarrow \mathrm{Cu(m)} + \mathrm{H^+(aq)} \qquad n = 1$$

$$E = E^\circ(\mathrm{Cu^+/Cu}) - \frac{RT}{F} \ln \frac{([\mathrm{H^+}]/c^\circ)}{(p_{\mathrm{H_2}}/p^\circ)^{\frac{1}{2}}\,([\mathrm{Cu^+}]/c^\circ)}$$

$$E^\circ_{\mathrm{cell}} = 0.518\ \mathrm{V} \qquad \text{spontaneous = conventional}$$

2

$$\mathrm{Pt(m)|H_2(g)|H^+(aq)||Fe^{2+}(aq)|Fe(m)}$$

$$\text{RHS} - 2\times\text{LHS} \quad \mathrm{Fe^{2+}(aq)} + \mathrm{H_2(g)} \longrightarrow 2\mathrm{H^+(aq)} + \mathrm{Fe(m)} \qquad n = 2$$

$$E = E^\circ(\mathrm{Fe^{2+}/Fe}) - \frac{RT}{2F} \ln \frac{([\mathrm{H^+}]/c^\circ)^2}{(p_{\mathrm{H_2}}/p^\circ)\,([\mathrm{Fe^{2+}}]/c^\circ)} \qquad (21.\mathrm{II})$$

$$E^\circ_{\mathrm{cell}} = -0.477\ \mathrm{V} \qquad \text{spontaneous = reverse of conventional}$$

3

$$\mathrm{Fe(m)|Fe^{2+}(aq)||H^+(aq)|H_2(g)|Pt(m)}$$

$$2\times\text{RHS} - \text{LHS} \quad 2\mathrm{H^+(aq)} + \mathrm{Fe(m)} \longrightarrow \mathrm{Fe^{2+}(aq)} + \mathrm{H_2(g)} \qquad n = 2$$

$$E = -E^\circ(\mathrm{Fe^{2+}/Fe}) - \frac{RT}{2F} \ln \frac{(p_{\mathrm{H_2}}/p^\circ)\left([\mathrm{Fe^{2+}}]/c^\circ\right)}{([\mathrm{H^+}]/c^\circ)^2}$$

$$E^\circ_{\mathrm{cell}} = 0.477\ \mathrm{V} \qquad \text{spontaneous = conventional}$$

4

$$\mathrm{Mn(m)|Mn^{2+}(aq)||Au^{3+}(aq)|Au(m)}$$

$$2\times\text{RHS} - 3\times\text{LHS} \quad 2\mathrm{Au^{3+}(aq)} + 3\mathrm{Mn(m)} \longrightarrow 2\mathrm{Au(m)} + 3\mathrm{Mn^{2+}(aq)} \qquad n = 6$$

$$E = E^\circ(\mathrm{Au^{3+}/Au}) - E^\circ(\mathrm{Mn^{2+}/Mn}) - \frac{RT}{6F} \ln \frac{\left([\mathrm{Mn^{2+}}]/c^\circ\right)^3}{([\mathrm{Au^{3+}}]/c^\circ)^2}$$

$$E^\circ_{\mathrm{cell}} = 1.498 - (-1.185) = 2.683\ \mathrm{V} \qquad \text{spontaneous = conventional}$$

5

$$\mathrm{Ag(m), AgCl(s)|Cl^-(aq)||H^+(aq)|H_2(g)|Pt(m)}$$

$$\text{RHS} - \text{LHS} \quad \mathrm{H^+(aq) + Ag(m) + Cl^-(aq) \longrightarrow AgCl(s) + \tfrac{1}{2}H_2(g)} \qquad n = 1$$

$$E = -E^\circ(\mathrm{AgCl/Cl^-}) - \frac{RT}{F}\ln\frac{(p_{\mathrm{H_2}}/p^\circ)^{\frac{1}{2}}}{([\mathrm{H^+}]/c^\circ)([\mathrm{Cl^-}]/c^\circ)}$$

$$E^\circ_{\text{cell}} = -0.2223\ \text{V} \qquad \text{spontaneous = reverse of conventional}$$

6

$$\mathrm{Ag(m), AgI(s)|I^-(aq)||Zn^{2+}(aq)|Zn(m)}$$

$$\text{RHS} - 2\times\text{LHS} \quad \mathrm{Zn^{2+}(aq) + 2Ag(m) + 2I^-(aq) \longrightarrow 2AgI(s) + Zn(m)} \qquad n = 2$$

$$E = E^\circ(\mathrm{Zn^{2+}/Zn}) - E^\circ(\mathrm{AgI/I^-}) - \frac{RT}{2F}\ln\frac{1}{([\mathrm{Zn^{2+}}]/c^\circ)([\mathrm{I^-}]/c^\circ)^2}$$

$$E^\circ_{\text{cell}} = -0.7618 - (-0.1522) = -0.6096\ \text{V} \qquad \text{spontaneous = reverse of conventional}$$

7

$$\mathrm{Ag(m), AgCl(s)|Cl^-(aq)_L||Cl^-(aq)_R|Hg_2Cl_2(s)|Hg(l)}$$

$$\text{RHS} - 2\times\text{LHS} \quad \mathrm{Hg_2Cl_2(s) + 2Ag(m) + 2Cl^-(aq)_L \longrightarrow 2Hg(l) + 2Cl^-(aq)_R + 2AgCl(s)} \qquad n = 2$$

$$E = E^\circ(\mathrm{Hg_2Cl_2/Cl^-}) - E^\circ(\mathrm{AgCl/Cl^-}) - \frac{RT}{2F}\ln\frac{\left([\mathrm{Cl_R^-}]/c^\circ\right)^2}{\left([\mathrm{Cl_L^-}]/c^\circ\right)^2}$$

$$E^\circ_{\text{cell}} = 0.2681 - (0.2223) = 0.0458\ \text{V} \qquad \text{spontaneous = conventional}$$

8

$$\mathrm{Pt(m)|H_2(g)|OH^-(aq)_L||OH^-(aq)_R|O_2(g)|Pt(m)}$$

$$\text{RHS} - 2\times\text{LHS} \quad \mathrm{O_2(g) + 2H_2(g) + 4OH^-(aq)_L \longrightarrow 4OH^-(aq)_R + 2H_2O(l)} \qquad n = 4$$

$$E = E^\circ(\mathrm{O_2/OH^-}) - E^\circ(\mathrm{H_2O/H_2, OH^-}) - \frac{RT}{4F}\ln\frac{\left([\mathrm{OH_R^-}]/c^\circ\right)^4}{(p_{\mathrm{O_2}}/p^\circ)(p_{\mathrm{H_2}}/p^\circ)^2\left([\mathrm{OH_L^-}]/c^\circ\right)^4}$$

$$E^\circ_{\text{cell}} = 0.401 - (-0.8277) = 1.2287\ \text{V} \qquad \text{spontaneous = conventional}$$

9

$$\mathrm{Pt(m)|O_2(g)|H^+(aq)_L||Mn^{2+}(aq), MnO_4^-(aq), H^+(aq)_R||Pt(m)}$$

$$4\times\text{RHS} - 5\times\text{LHS}$$

$$\mathrm{4MnO_4^-(aq) + 32H^+(aq)_R \longrightarrow 4Mn^{2+}(aq) + 6H_2O + 5O_2(g) + 20\,H^+(aq)_L} \qquad n = 20$$

$$E = E^\circ(\mathrm{MnO_4^-/Mn^{2+}}) - E^\circ(\mathrm{O_2/H_2O}) - \frac{RT}{20F}\ln\frac{\left([\mathrm{Mn^{2+}}]/c^\circ\right)^4(p_{\mathrm{O_2}}/p^\circ)^5\left([\mathrm{H_L^+}]/c^\circ\right)^{20}}{\left([\mathrm{MnO_4^-}]/c^\circ\right)^4\left([\mathrm{H_R^+}]/c^\circ\right)^{32}}$$

$$E^\circ_{\text{cell}} = 1.507 - (1.229) = 0.278\ \text{V} \qquad \text{spontaneous = conventional}$$

10

$$\mathrm{Pt(m)|H_2(g)|OH^-(aq)_L||OH^-(aq)_R|PbO(s), Pb(m)}$$

$$\text{RHS} - \text{LHS} \quad \mathrm{PbO(s) + H_2(g) + 2OH^-(aq)_L \longrightarrow Pb(m) + 2OH^-(aq)_R + H_2O(l)} \qquad n = 2$$

$$E = E^\circ(\mathrm{PbO/OH^-}) - E^\circ(\mathrm{H_2O/OH^-}) - \frac{RT}{2F}\ln\frac{\left([\mathrm{OH_R^-}]/c^\circ\right)^2}{\left([\mathrm{OH_L^-}]/c^\circ\right)^2(p_{\mathrm{H_2}}/p^\circ)}$$

$$E^\circ_{\text{cell}} = -0.580 - (-0.8277) = 0.2477\ \text{V} \qquad \text{spontaneous = conventional}$$

21.2

The same considerations apply here as in the previous exercise. It is up to you how many electrons you choose to include in the individual half-cell reactions, provided that when you compute the cell reaction you make sure that you have the same number of electrons in each half-cell reaction.

1

$$\text{RHS} \quad Mg^{2+}(aq) + 2e^- \longrightarrow Mg(m)$$

$$\text{LHS} \quad Zn^{2+}(aq) + 2e^- \longrightarrow Zn(m)$$

$$\text{RHS} - \text{LHS} \quad Mg^{2+}(aq) + Zn(m) \longrightarrow Mg(m) + Zn^{2+}(aq) \quad n = 2$$

$$E = E^\circ(Mg^{2+}/Mg) - E^\circ(Zn^{2+}/Zn) - \frac{RT}{2F} \ln \frac{\left([Zn^{2+}]/c^\circ\right)}{([Mg^{2+}]/c^\circ)}$$

$E^\circ_{\text{cell}} = -2.372 - (-0.7618) = -1.6102$ V spontaneous = reverse of conventional

2

$$\text{RHS} \quad Fe^{3+}(aq) + 3e^- \longrightarrow Fe(m)$$

$$\text{LHS} \quad 2H^+(aq) + 2e^- \longrightarrow H_2(g)$$

$$2 \times \text{RHS} - 3 \times \text{LHS} \quad 2Fe^{3+}(aq) + 3H_2(g) \longrightarrow 2Fe(m) + 6H^+(aq) \quad n = 6$$

$$E = E^\circ(Fe^{3+}/Fe) - \frac{RT}{6F} \ln \frac{([H^+]/c^\circ)^6}{([Fe^{3+}]/c^\circ)^2\,(p_{H_2}/p^\circ)^3}$$

$E^\circ_{\text{cell}} = -0.037$ V spontaneous = reverse of conventional

3

$$\text{RHS} \quad Cu^{2+}(aq) + 2e^- \longrightarrow Cu(m)$$

$$\text{LHS} \quad Sn^{4+}(aq) + 2e^- \longrightarrow Sn^{2+}(aq)$$

$$\text{RHS} - \text{LHS} \quad Cu^{2+}(aq) + Sn^{2+}(aq) \longrightarrow Cu(m) + Sn^{4+}(aq) \quad n = 2$$

$$E = E^\circ(Cu^{2+}/Cu) - E^\circ(Sn^{4+}/Sn^{2+}) - \frac{RT}{2F} \ln \frac{\left([Sn^{4+}]/c^\circ\right)}{([Cu^{2+}]/c^\circ)\,([Sn^{2+}]/c^\circ)}$$

$E^\circ_{\text{cell}} = 0.3419 - (0.151) = 0.1909$ V spontaneous = conventional

4

$$\text{RHS} \quad 2H^+(aq) + 2e^- \longrightarrow H_2(g)$$

$$\text{LHS} \quad AgI(s) + e^- \longrightarrow Ag(m) + I^-(aq)$$

$$\text{RHS} - 2 \times \text{LHS} \quad 2H^+(aq) + 2Ag(m) + 2I^-(aq) \longrightarrow 2AgI(s) + H_2(g) \quad n = 2$$

$$E = -E^\circ(AgI/I^-) - \frac{RT}{2F} \ln \frac{(p_{H_2}/p^\circ)}{([H^+]/c^\circ)^2\,([I^-]/c^\circ)^2}$$

$E^\circ_{\text{cell}} = -(-0.1522) = 0.1522$ V spontaneous = conventional

5 $H^+(aq)$ solution is the same on the RHS and LHS

$$\text{RHS} \quad O_2(g) + 4H^+(aq) + 4e^- \longrightarrow 2H_2O(l)$$

$$\text{LHS} \quad 2H^+(aq) + 2e^- \longrightarrow H_2(g)$$

$$\text{RHS} - 2 \times \text{LHS} \quad O_2(g) + 2H_2(g) \longrightarrow 2H_2O(l) \quad n = 4$$

$$E = E^\circ(O_2/H_2O) - \frac{RT}{4F} \ln \frac{1}{(p_{O_2}/p^\circ)\,(p_{H_2}/p^\circ)^2}$$

$E^\circ_{\text{cell}} = 1.229$ V spontaneous = conventional

6 OH^-(aq) solution is the same on the RHS and LHS.

$$\text{RHS}\quad O_2(g) + 2H_2O(l) + 4e^- \longrightarrow 4OH^-(aq)$$

$$\text{LHS}\quad 2H_2O(l) + 2e^- \longrightarrow H_2(g) + 2OH^-(aq)$$

$$\text{RHS} - 2\times\text{LHS}\quad O_2(g) + 2H_2(g) \longrightarrow 2H_2O(l)\quad n = 4$$

$$E = E^\circ(O_2/OH^-) - E^\circ(H_2/OH^-) - \frac{RT}{4F}\ln\frac{1}{(p_{O_2}/p^\circ)(p_{H_2}/p^\circ)^2}$$

$$E^\circ_{\text{cell}} = 0.401 - (-0.8277) = 1.2287\ \text{V}\qquad \text{spontaneous = conventional}$$

7

$$\text{RHS}\quad Br_2(aq) + 2e^- \longrightarrow 2Br^-(aq)$$

$$\text{LHS}\quad AgBr(s) + e^- \longrightarrow Ag(m) + Br^-(aq)$$

$$\text{RHS} - 2\times\text{LHS}\quad Br_2(aq) + 2Ag(m) \longrightarrow 2AgBr(s)\quad n = 2$$

$$E = E^\circ(Br_2/Br^-) - E^\circ(AgBr/Br^-) - \frac{RT}{2F}\ln\frac{1}{([Br_2]/c^\circ)}$$

$$E^\circ_{\text{cell}} = 1.0873 - (0.0713) = 1.016\ \text{V}\qquad \text{spontaneous = conventional}$$

8

$$\text{RHS}\quad 2H_2O(l) + 2e^- \longrightarrow 2OH^-(aq) + H_2(g)$$

$$\text{LHS}\quad PbO(s) + H_2O(l) + 2e^- \longrightarrow Pb(m) + 2OH^-(aq)$$

$$\text{RHS} - \text{LHS}\quad H_2O(l) + Pb(m) \longrightarrow PbO(s) + H_2(g)\quad n = 2$$

$$E = E^\circ(H_2O/H_2) - E^\circ(PbO/OH^-) - \frac{RT}{2F}\ln(p_{H_2}/p^\circ)$$

$$E^\circ_{\text{cell}} = -0.8277 - (-0.580) = -0.2477\ \text{V}\qquad \text{spontaneous = reverse of conventional}$$

9

$$\text{RHS}\quad AgCl(s) + e^- \longrightarrow Ag(m) + Cl^-(aq)$$

$$\text{LHS}\quad Cr_2O_7^{2-}(aq) + 14H^+(aq) + 6e^- \longrightarrow 2Cr^{3+}(aq) + 7H_2O(l)$$

$$6\times\text{RHS} - \text{LHS}\quad 6AgCl(s) + 2Cr^{3+}(aq) + 7H_2O(l) \rightarrow 6Ag(m) + 6Cl^-(aq) + Cr_2O_7^{2-}(aq) + 14H^+(aq)$$

$$n = 6$$

$$E = E^\circ(AgCl/Cl^-) - E^\circ(Cr_2O_7^{2-}/Cr^{3+}) - \frac{RT}{6F}\ln\frac{([Cl^-]/c^\circ)^6\left([Cr_2O_7^{2-}]/c^\circ\right)([H^+]/c^\circ)^{14}}{([Cr^{3+}]/c^\circ)^2}$$

$$E^\circ_{\text{cell}} = 0.2223 - (1.232) = -1.0097\ \text{V}\qquad \text{spontaneous = reverse of conventional}$$

10

$$\text{RHS}\quad AgCl(s) + e^- \longrightarrow Ag(m) + Cl^-(aq)_R$$

$$\text{LHS}\quad AgCl(s) + e^- \longrightarrow Ag(m) + Cl^-(aq)_L$$

$$\text{RHS} - \text{LHS}\quad Cl^-(aq)_L \longrightarrow Cl^-(aq)_R\quad n = 1$$

$$E = -\frac{RT}{F}\ln\frac{c_R}{c_L}$$

$$E^\circ_{\text{cell}} = 0.00\ \text{V}$$

When the concentration in the LHS and RHS are the same (as they are under standard conditions) then the cell is already at equilibrium and so there is no spontaneous process. This is exactly what is implied by $E^\circ = 0$ and hence $\Delta_r G_{\text{cell}} = 0$.

21.3

Once again, the choice of number of electrons in each half-cell reaction is entirely up to you. This has no bearing on the direction of spontaneous cell reaction or the calculation of E°_{cell}. The trick with these problems is to identify which are the oxidized species on the RHS and LHS, and then construct the cell from this information.

1

$$\text{Zn(m)|Zn}^{2+}\text{(aq)||Cu}^{2+}\text{(aq)|Cu(m)}$$

$$\text{RHS} \quad \text{Cu}^{2+}\text{(aq)} + 2\text{e}^- \longrightarrow \text{Cu(m)}$$

$$\text{LHS} \quad \text{Zn}^{2+}\text{(aq)} + 2\text{e}^- \longrightarrow \text{Zn(m)}$$

$$E^\circ_{\text{cell}} = E^\circ(\text{Cu}^{2+}/\text{Cu}) - E^\circ(\text{Zn}^{2+}/\text{Zn}) = 0.3419 - (-0.7618) = 1.1037 \text{ V}$$

spontaneous = conventional

2

$$\text{Ag(m)|Ag}^{+}\text{(aq)||Sn}^{4+}\text{(aq), Sn}^{2+}\text{(aq)|Pt(m)}$$

$$\text{RHS} \quad \text{Sn}^{4+}\text{(aq)} + 2\text{e}^- \longrightarrow \text{Sn}^{2+}\text{(aq)}$$

$$\text{LHS} \quad \text{Ag}^{+}\text{(aq)} + \text{e}^- \longrightarrow \text{Ag(m)}$$

$$E^\circ_{\text{cell}} = E^\circ(\text{Sn}^{4+}/\text{Sn}^{2+}) - E^\circ(\text{Ag}^{+}/\text{Ag}) = 0.151 - (0.7996) = -0.6486 \text{ V}$$

spontaneous = reverse of conventional

3

$$\text{Pt(m)|H}_2\text{(g)|H}^{+}\text{(aq), Cl}^{-}\text{(aq)|AgCl(s)|Ag(m)}$$

$$\text{RHS} \quad \text{AgCl(s)} + \text{e}^- \longrightarrow \text{Ag(m)} + \text{Cl}^-\text{(aq)}$$

$$\text{LHS} \quad 2\text{H}^{+}\text{(aq)} + 2\text{e}^- \longrightarrow \text{H}_2\text{(g)}$$

$$E^\circ_{\text{cell}} = E^\circ(\text{AgCl/Cl}^-) - E^\circ(\text{H}^+/\text{H}_2) = 0.2223 - (0.000) = 0.2223 \text{ V}$$

spontaneous = conventional

4 This is just the reverse of 3

$$\text{Ag(m), AgCl(s)|Cl}^{-}\text{(aq), H}^{+}\text{(aq)|H}_2\text{(g)|Pt(m)}$$

$$\text{RHS} \quad 2\text{H}^{+}\text{(aq)} + 2\text{e}^- \longrightarrow \text{H}_2\text{(g)}$$

$$\text{LHS} \quad \text{AgCl(s)} + \text{e}^- \longrightarrow \text{Ag(m)} + \text{Cl}^-\text{(aq)}$$

$$E^\circ_{\text{cell}} = E^\circ(\text{H}^+/\text{H}_2) - E^\circ(\text{AgCl/Cl}^-) = 0.000 - (0.2223) = -0.2223 \text{ V}$$

spontaneous = reverse of conventional

5 This is just twice the reaction shown in 3. The cell is the same, as is the standard potential, and the direction of the spontaneous reaction.

6 The Cl^-(aq) solution must be common to both sides for it to cancel out in the cell reaction

$$\text{Hg(l)|Hg}_2\text{Cl}_2\text{(s)|Cl}^{-}\text{(aq)|AgCl(s), Ag(m)}$$

$$\text{RHS} \quad \text{AgCl(s)} + \text{e}^- \longrightarrow \text{Ag(m)} + \text{Cl}^-\text{(aq)}$$

$$\text{LHS} \quad \text{Hg}_2\text{Cl}_2\text{(s)} + 2\text{e}^- \longrightarrow 2\text{Hg(l)} + 2\text{Cl}^-\text{(aq)}$$

$$E^\circ_{\text{cell}} = E^\circ(\text{AgCl/Cl}^-) - E^\circ(\text{Hg}_2\text{Cl}_2/\text{Cl}^-) = 0.2223 - (0.2681) = -0.0458 \text{ V}$$

spontaneous = reverse of conventional

7

$$\mathrm{Ag(m), AgCl(s)|Cl^-(aq), H^+(aq)|O_2(g)|Pt(m)}$$

$$\text{RHS} \quad \mathrm{O_2(g) + 4H^+(aq) + 4e^- \longrightarrow 2H_2O(l)}$$

$$\text{LHS} \quad \mathrm{AgCl(s) + e^- \longrightarrow Ag(m) + Cl^-(aq)}$$

$$E^\circ_{\text{cell}} = E^\circ(\mathrm{O_2/H_2O}) - E^\circ(\mathrm{AgCl/Cl^-}) = 1.229 - (0.2223) = 1.0067\ \text{V}$$

spontaneous = conventional

8 The OH^-(aq) solution must be common to both sides for it to cancel out in the cell reaction

$$\mathrm{Pb(m), PbO(s)|OH^-(aq)|H_2(g)|Pt(m)}$$

$$\text{RHS} \quad \mathrm{2H_2O(l) + 2e^- \longrightarrow 2OH^-(aq) + H_2(g)}$$

$$\text{LHS} \quad \mathrm{PbO(s) + H_2O(l) + 2e^- \longrightarrow Pb(m) + 2OH^-(aq)}$$

$$E^\circ_{\text{cell}} = E^\circ(\mathrm{H_2O/OH^-, H_2}) - E^\circ(\mathrm{PbO/OH^-}) = -0.8277 - (-0.580) = -0.2477\ \text{V}$$

spontaneous = reverse of conventional

9

$$\mathrm{Ag(m)|Ag^+(aq), Br^-(aq)|AgBr(s), Ag(m)}$$

$$\text{RHS} \quad \mathrm{AgBr(s) + e^- \longrightarrow Ag(m) + Br^-(aq)}$$

$$\text{LHS} \quad \mathrm{Ag^+(aq) + e^- \longrightarrow Ag(m)}$$

$$E^\circ_{\text{cell}} = E^\circ(\mathrm{AgBr/Br^-}) - E^\circ(\mathrm{Ag^+/Ag}) = 0.0713 - (0.7996) = -0.7283\ \text{V}$$

spontaneous = reverse of conventional

10

$$\mathrm{Al(m)|Al^{3+}(aq)||Ce^{4+}(aq), Ce^{3+}(aq)|Pt(m)}$$

$$\text{RHS} \quad \mathrm{Ce^{4+}(aq) + e^- \longrightarrow Ce^{3+}(aq)}$$

$$\text{LHS} \quad \mathrm{Al^{3+}(aq) + 3e^- \longrightarrow Al(m)}$$

$$E^\circ_{\text{cell}} = E^\circ(\mathrm{Ce^{4+}/Ce^{3+}}) - E^\circ(\mathrm{Al^{3+}/Al}) = 1.72 - (-1.662) = 3.382\ \text{V}$$

spontaneous = conventional

21.4

(a)

$$\text{RHS} \quad \mathrm{Cd^{2+}(aq) + 2e^- \longrightarrow Cd(m)}$$

$$\text{LHS} \quad \mathrm{Fe^{2+}(aq) + 2e^- \longrightarrow Fe(m)}$$

$$\text{conventional} = \text{RHS} - \text{LHS} \quad \mathrm{Cd^{2+}(aq) + Fe(m) \longrightarrow Cd(m) + Fe^{2+}(aq)}$$

$$\begin{aligned} E &= E^\circ(\mathrm{Cd^{2+}/Cd}) - E^\circ(\mathrm{Fe^{2+}/Fe}) - \frac{RT}{2F}\ln\frac{\left([\mathrm{Fe^{2+}}]/c^\circ\right)}{\left([\mathrm{Cd^{2+}}]/c^\circ\right)} \\ &= E^\circ(\mathrm{Cd^{2+}/Cd}) - E^\circ(\mathrm{Fe^{2+}/Fe}) - \frac{RT}{2F}\ln\frac{c_2}{c_1} \end{aligned}$$

(b)

$$E^\circ_{\text{cell}} = E^\circ(\text{Cd}^{2+}/\text{Cd}) - E^\circ(\text{Fe}^{2+}/\text{Fe}) = -0.4030 - (-0.447) = 0.044 \text{ V}$$

The spontaneous cell reaction is therefore the same as the conventional cell reaction.

(c)

$$\begin{aligned}
0 &= E^\circ(\text{Cd}^{2+}/\text{Cd}) - E^\circ(\text{Fe}^{2+}/\text{Fe}) - \frac{RT}{2F}\ln\left(\frac{c_2}{c_1}\right) \\
&= 0.044 - \frac{RT}{2F}\ln\left(\frac{c_2}{c_1}\right) \\
\text{hence } \ln\left(\frac{c_2}{c_1}\right) &= \frac{0.044 \times 2F}{RT} \\
\ln\left(\frac{c_2}{c_1}\right) &= \frac{0.044 \times 2 \times 96485}{8.3145 \times 298} \\
\ln\left(\frac{c_2}{c_1}\right) &= 3.427 \\
\text{hence } \frac{c_2}{c_1} &= 30.8
\end{aligned}$$

(d) For the spontaneous cell reaction to be *opposite* to that in (b), we need the cell potential to be *negative*. This is achieved by making $c_2/c_1 > 30.8$, since this will make the $(-RT/2F)\ln(c_2/c_1)$ term more negative that 0.044 (which is E°_{cell}), and hence make the cell potential negative.

21.5

Following section 21.7 on page 811:

(a) $E^\circ(\text{Cu}^{2+}/\text{Cu}) = +0.3419$ V and $E^\circ(\text{Fe}^{3+}/\text{Fe}^{2+}) = +0.771$ V. Since $E^\circ(\text{Fe}^{3+}/\text{Fe}^{2+}) > E^\circ(\text{Cu}^{2+}/\text{Cu})$ it follows that Fe^{3+} is more powerfully oxidizing than Cu^{2+}. Thus Fe^{3+} will oxidize Cu to Cu^{2+}, in the process being reduced itself to Fe^{2+}.

Note that since $E^\circ(\text{Fe}^{3+}/\text{Fe}) = -0.037$ V, the reaction in which Cu is oxidized and in which the iron goes to Fe (rather than Fe^{2+}) is *not* thermodynamically feasible.

(b) $E^\circ(\text{Cl}_2/\text{Cl}^-) = +1.358$ V and $E^\circ(\text{Fe}^{3+}/\text{Fe}^{2+}) = +0.771$ V. This means that Cl_2 is more powerfully oxidizing than Fe^{3+}, and so Cl_2 will oxidize Fe^{2+} to Fe^{3+}, in the process being reduced itself to Cl^-.

(c) For the oxygen we need the half-cell $O_2(g) + 4H^+(aq) + 4e^- \longrightarrow 2H_2O(l)$. $E^\circ(\text{O}_2/\text{H}_2\text{O}) = +1.229$ V and $E^\circ(\text{Fe}^{3+}/\text{Fe}^{2+}) = +0.771$ V. This means that (in acid conditions) O_2 is more powerfully oxidizing that Fe^{3+}, and so Fe^{3+} *will not* oxidize H_2O to O_2. On the contrary, O_2 will oxidize Fe^{2+} to Fe^{3+}.

(d) $E^\circ(\text{Cu}^{2+}/\text{Cu}^+) = +0.161$ V and $E^\circ(\text{Ag}^+/\text{Ag}) = +0.7996$ V. Since $E^\circ(\text{Ag}^+/\text{Ag}) > E^\circ(\text{Cu}^{2+}/\text{Cu}^+)$ it follows that Ag^+ is more powerfully oxidizing than Cu^{2+}. In terms of reduction, since $E^\circ(\text{Cu}^{2+}/\text{Cu}^+) < E^\circ(\text{Ag}^+/\text{Ag})$ it follows that Cu^+ is more powerfully reducing that Ag. Thus Ag will *not* reduce Cu^{2+} to Cu^+.

(e) $E^\circ(\text{Fe}^{3+}/\text{Fe}) = -0.037$ V and $E^\circ(\text{Zn}^{2+}/\text{Zn}) = -0.7618$ V. Since $E^\circ(\text{Zn}^{2+}/\text{Zn}) < E^\circ(\text{Fe}^{3+}/\text{Fe})$ it follows that Zn is more powerfully reducing that Fe. Thus Zn will reduce Fe^{3+} to Fe.

(f) $E^\circ(\text{Mn}^{2+}/\text{Mn}) = -1.185$ V and $E^\circ(\text{Tl}^+/\text{Tl}) = -0.336$ V. Since $E^\circ(\text{Mn}^{2+}/\text{Mn}) < E^\circ(\text{Tl}^+/\text{Tl}))$ it follows that Mn is more powerfully reducing than Tl. Thus Tl will *not* reduce Mn^{2+} to Mn.

21.6

(a) $n = 1$ and $E^\circ(H^+/H_2) = 0.00$, so

$$E = -\frac{RT}{F}\ln\left(\frac{(p_{H_2}/p^\circ)^{\frac{1}{2}}}{([H^+]/c^\circ)}\right).$$

(b) If $p_{H_2} = 1$ bar, and recalling that $p^\circ = 1$ bar, the term (p_{H_2}/p°) is $= 1$. Using $-\ln(x) \equiv \ln(1/x)$ gives

$$E = \frac{RT}{F}\ln([H^+]/c^\circ).$$

If we assume that $[H^+]$ is in mol dm^{-3}, then as $c^\circ = 1$ mol dm^{-3}, we can write

$$E = \frac{RT}{F}\ln[H^+]. \qquad (21.III)$$

(c) By definition $\text{pH} = -\log[H^+]$. Using the hint we have

$$\ln[H^+] \equiv \ln 10 \times \log[H^+]$$

which means that

$$\ln[H^+] \equiv \ln 10 \times (-\text{pH}).$$

Putting this into Eq. 21.III we have

$$E = -\ln 10 \times \frac{RT}{F} \times \text{pH}.$$

(d) As the pH increases, this last equation tells us that the half-cell potential *decreases*. In other words, H_2 becomes a more powerful reducing agent as the pH is increased. As a result, the reduction of H^+ becomes more difficult as the pH is increased.

21.7

(a)

$$\text{Ag(m)}|\text{Ag}^+\text{(aq)}, \text{Cl}^-\text{(aq)}|\text{AgCl(s)}, \text{Ag(m)}$$

$$\text{RHS} \quad \text{AgCl(s)} + e^- \longrightarrow \text{Ag(m)} + \text{Cl}^-\text{(aq)}$$

$$\text{LHS} \quad \text{Ag}^+\text{(aq)} + e^- \longrightarrow \text{Ag(m)}$$

$$\text{conventional} = \text{RHS} - \text{LHS} \quad \text{AgCl(s)} \longrightarrow \text{Ag}^+\text{(aq)} + \text{Cl}^-\text{(aq)}$$

(b)

$$E^\circ_{\text{cell}} = E^\circ(\text{AgCl/Cl}^-) - E^\circ(\text{Ag}^+/\text{Ag}) = 0.2223 - (0.7996) = -0.5773 \text{ V}$$

$$\begin{aligned}\Delta_r G^\circ_{\text{cell}} &= -nFE^\circ \\ &= -1 \times 96485 \times (-0.5773) \\ &= 5.57 \times 10^4 \text{ J mol}^{-1} \\ &= 55.7 \text{ kJ mol}^{-1}.\end{aligned}$$

(c) The solubility product is the equilibrium constant for the reaction

$$\mathrm{AgCl(s)} \longrightarrow \mathrm{Ag^+(aq)} + \mathrm{Cl^-(aq)} \quad K = ([\mathrm{Ag^+}]/c^\circ)([\mathrm{Cl^-}]/c^\circ).$$

We have already worked out $\Delta_r G^\circ$ for this reaction, as it is the conventional cell reaction. Given that $\Delta_r G^\circ = -RT \ln K$ it follows that

$$\begin{aligned} K &= \exp\left(\frac{-\Delta_r G^\circ_{\mathrm{cell}}}{RT}\right) \\ &= \exp\left(\frac{-55.7 \times 10^3}{8.3145 \times 298}\right) \\ &= 1.73 \times 10^{-10}. \end{aligned}$$

If all that is happening is that AgCl is dissolving to Ag^+ and Cl^-, it follows that $[\mathrm{Ag^+}] = [\mathrm{Cl^-}]$. We therefore have

$$K = ([\mathrm{Ag^+}]/c^\circ)^2.$$

It follows that $[\mathrm{Ag^+}] = c^\circ \times \sqrt{K}$, which is 1.32×10^{-5} mol dm^{-3} (recall that $c^\circ = 1$ mol dm^{-3}).

21.8

(a)

$$2 \times \mathrm{RHS} - \mathrm{LHS} \quad 2\mathrm{H^+(aq)} + \mathrm{Mg(m)} \longrightarrow \mathrm{H_2(g)} + \mathrm{Mg^{2+}(aq)}$$

(b) Using the results in section 21.2.3 on page 798 and noting that $n = 2$

$$\begin{aligned} \Delta_r G^\circ_{\mathrm{cell}} &= -nFE^\circ \\ &= -2 \times 96485 \times (2.360) \\ &= -4.554 \times 10^5 \ \mathrm{J\ mol^{-1}} \\ &= -455.4 \ \mathrm{kJ\ mol^{-1}}. \end{aligned}$$

$$\begin{aligned} \Delta_r S^\circ_{\mathrm{cell}} &= nF\frac{\mathrm{d}E^\circ}{\mathrm{d}T} \\ &= 2 \times 96485 \times \left(-1.99 \times 10^{-4}\right) \\ &= -38.4 \ \mathrm{J\ K^{-1}\ mol^{-1}}. \end{aligned}$$

Since $\Delta_r G^\circ_{\mathrm{cell}} = \Delta_r H^\circ_{\mathrm{cell}} - T\Delta_r S^\circ_{\mathrm{cell}}$

$$\begin{aligned} \Delta_r H^\circ_{\mathrm{cell}} &= \Delta_r G^\circ_{\mathrm{cell}} + T\Delta_r S^\circ_{\mathrm{cell}} \\ &= -455.4 \times 10^3 + 298 \times (-38.4) \\ &= -4.668 \times 10^5 \ \mathrm{J\ mol^{-1}} \\ &= -466.8 \ \mathrm{kJ\ mol^{-1}}. \end{aligned}$$

(c) From the conventional cell reaction it follows that

$$\Delta_r G^\circ_{\mathrm{cell}} = \underbrace{\Delta_f G^\circ(\mathrm{H_2(g)})}_{=0} + \Delta_f G^\circ(\mathrm{Mg^{2+}(aq)}) - 2\,\underbrace{\Delta_f G^\circ(\mathrm{H^+(aq)})}_{=0} - \underbrace{\Delta_f G^\circ(\mathrm{Mg(m)})}_{=0}$$

The first and last terms on the right are zero as these are the standard Gibbs energies of formation of elements in their reference phases. The third term is zero by convention (see section 21.9.2 on page 817). It therefore follows that

$$\Delta_f G^\circ(\mathrm{Mg^{2+}(aq)}) = -455.4 \ \mathrm{kJ\ mol^{-1}}.$$

Similarly

$$\Delta_r H^\circ_{cell} = \underbrace{\Delta_f H^\circ(H_2(g))}_{=0} + \Delta_f H^\circ(Mg^{2+}(aq)) - 2\,\underbrace{\Delta_f H^\circ(H^+(aq))}_{=0} - \underbrace{\Delta_f H^\circ(Mg(m))}_{=0}$$

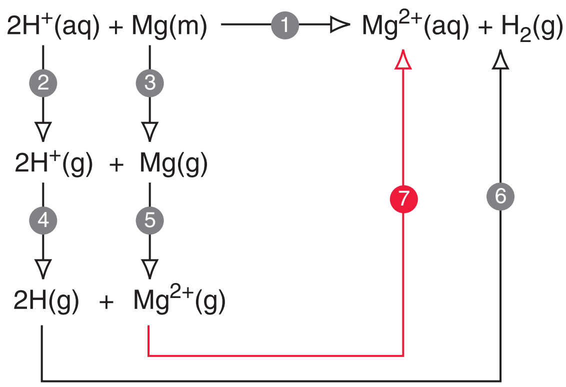
hence

$$\Delta_f H^\circ(Mg^{2+}(aq)) = -466.8 \text{ kJ mol}^{-1}.$$

Since $\Delta_f G^\circ = \Delta_f H^\circ - T\Delta_f S^\circ$

$$\begin{aligned}\Delta_f S^\circ(Mg^{2+}(aq)) &= \frac{1}{T}\left(\Delta_f H^\circ(Mg^{2+}(aq)) - \Delta_f G^\circ(Mg^{2+}(aq))\right)\\ &= \frac{1}{298}\left(-466.8 \times 10^3 - (-455.4 \times 10^3)\right)\\ &= -38.3 \text{ J K}^{-1} \text{ mol}^{-1}.\end{aligned}$$

(d) Following the example in the text, we have



Looking at the steps one by one:

1 This is the cell reaction, $\Delta_r H^\circ_{cell} = -466.8$ kJ mol^{-1}.
2 This is two times *minus* the enthalpy of hydration of $H^+(g)$ i.e. $2 \times 1110 = 2220$ kJ mol^{-1}.
3 This is the enthalpy of atomization of Mg metal, 147.1 kJ mol^{-1}.
4 This is *minus* the enthalpy change on ionizing two H(g) atoms, i.e. $2 \times -1312 = -2624$ kJ mol^{-1}.
5 This is the enthalpy change on doubly ionizing Mg(g), 1450.7 kJ mol^{-1}.
6 This is *minus* the enthalpy change on dissociating $H_2(g)$ to 2H(g), -456 kJ mol^{-1}.
7 Is what we require

Since $\Delta_r H^\circ(1) = \Delta_r H^\circ(2) + \Delta_r H^\circ(3) + \Delta_r H^\circ(4) + \Delta_r H^\circ(5) + \Delta_r H^\circ(6) + \Delta_r H^\circ(7)$, it follows that

$$\begin{aligned}\Delta_r H^\circ(7) &= \Delta_r H^\circ(1) - \Delta_r H^\circ(2) - \Delta_r H^\circ(3) - \Delta_r H^\circ(4) - \Delta_r H^\circ(5) - \Delta_r H^\circ(6)\\ &= (-466.8) - (2220) - (147.1) - (-2624) - (1450.7) - (-456)\\ &= -1205 \text{ kJ mol}^{-1}.\end{aligned}$$

This is the absolute enthalpy of hydration as we have *not* assumed that $\Delta_f H^\circ(H^+(aq)) = 0$ anywhere in this calculation.

21.9

We will need Eq. 21.26 and Eq. 21.27 on page 821:

$$\Delta_f G^\circ(M^{m+}) = \Delta_f G^\circ(M^{(m-n)+}) + nFE^\circ(M^{m+}/M^{(m-n)+})$$

$$\Delta_f G^\circ(M^{m+}) = mFE^\circ(M^{m+}/M).$$

The latter equation applies when the reduced form in the couple is simply the metal.
Since it is usual to plot $\Delta_f G^\circ(M^{m+})/F$, these equations are more usefully expressed as

$$\Delta_f G^\circ(M^{m+})/F = \Delta_f G^\circ(M^{(m-n)+})/F + nE^\circ(M^{m+}/M^{(m-n)+}) \quad (21.IV)$$

$$\Delta_f G^\circ(M^{m+})/F = mE^\circ(M^{m+}/M). \quad (21.V)$$

From the given value of $E^\circ(Co^{2+}/Co)$ we can deduce immediately from Eq. 21.V (with $m = 2$) that

$$\Delta_f G^\circ(Co^{2+})/F = 2 \times (-0.282) = -0.564 \text{ V}.$$

Using this and the given value of $E^\circ(Co^{3+}/Co^{2+})$ in Eq. 21.IV, and noting that $m = 3$ and $n = 1$, gives

$$\begin{aligned} \Delta_f G^\circ(M^{m+})/F &= \Delta_f G^\circ(M^{(m-n)+})/F + nE^\circ(M^{m+}/M^{(m-n)+}) \\ \text{hence} \quad \Delta_f G^\circ(Co^{3+})/F &= \Delta_f G^\circ(Co^{2+})/F + 1 \times E^\circ(Co^{3+}/Co^{2+}) \\ &= -0.564 + 1 \times (1.92) \\ &= 1.356 \text{ V}. \end{aligned}$$

For the Co(IV) oxide we need to take into account that H_2O is formed. Imagine the following cell:

$$\text{RHS } CoO_2(s) + 4H^+(aq) + e^- \longrightarrow Co^{3+}(aq) + 2H_2O(l)$$

$$\text{LHS } H^+(aq) + e^- \longrightarrow \tfrac{1}{2}H_2(g)$$

for which the conventional cell reaction is

$$CoO_2(s) + 3H^+ + \tfrac{1}{2}H_2(g) \longrightarrow Co^{3+}(aq) + 2H_2O(l)$$

$\Delta_r G^\circ_{cell}$ is therefore

$$\Delta_r G^\circ_{cell} = \Delta_f G^\circ(Co^{3+}(aq)) + 2\Delta_f G^\circ(H_2O(l)) - \Delta_f G^\circ(CoO_2(s)) - \tfrac{1}{2}\underbrace{\Delta_f G^\circ(H_2(g))}_{=0} - 3\underbrace{\Delta_f G^\circ(H^+(aq))}_{=0},$$

note the two terms which are zero by convention (see section 21.9.2 on page 817)
Since $\Delta_r G^\circ_{cell} = -1 \times F \times E^\circ(CoO_2/Co^{3+})$ we can write this last equation as

$$-FE^\circ(CoO_2/Co^{3+}) = \Delta_f G^\circ(Co^{3+}(aq)) + 2\Delta_f G^\circ(H_2O(l)) - \Delta_f G^\circ(CoO_2(s))$$

Rewriting this gives

$$\Delta_f G^\circ(CoO_2(s)) - 2\Delta_f G^\circ(H_2O(l)) = \Delta_f G^\circ(Co^{3+}(aq)) + FE^\circ(CoO_2/Co^{3+}),$$

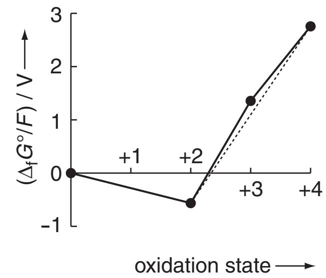
dividing by F gives

$$(\Delta_f G^\circ(CoO_2(s)) - 2\Delta_f G^\circ(H_2O(l)))/F = \Delta_f G^\circ(Co^{3+}(aq))/F + E^\circ(CoO_2/Co^{3+}).$$

Putting in the value for $\Delta_f G^\circ(Co^{3+}(aq))/F$ that we have already obtained, along with the given value for $E^\circ(CoO_2/Co^{3+})$ we obtain

$$(\Delta_f G^\circ(CoO_2(s)) - 2\Delta_f G^\circ(H_2O(l)))/F = 1.356 + 1.4 = 2.756 \text{ V}.$$

Of course all this boils down to simply ignoring the water involved and using Eq. 21.IV directly with $m = 4$ and $n = 1$.
The oxidation state diagram is thus

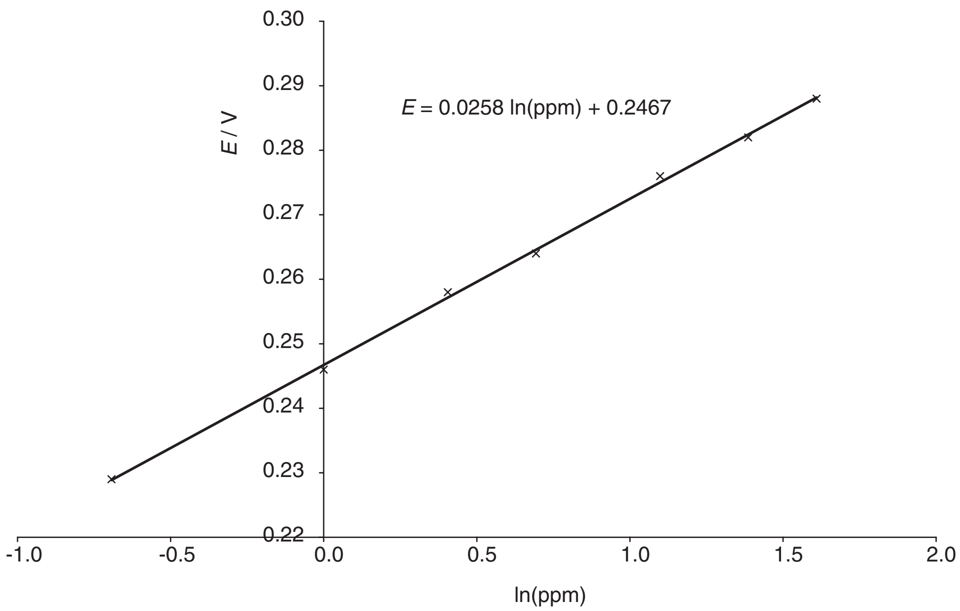


From this we note that the +2 oxidation state is clearly the most stable of the oxidation states. With its high $\Delta_f G°/F$ value Co(IV) is a strong oxidizing agent. Finally, the dotted line shows that (under acid conditions) Co(III) is predicted to be unstable (just) with respect to disproportionation to Co(II) and Co(IV).

21.10

We will plot the cell potential E against the (natural) logarithm of the concentration in ppm.

E / V	0.288	0.282	0.276	0.264	0.258	0.246	0.229
ppm F^-	5	4	3	2	1.5	1	0.5
ln (ppm)	1.609	1.386	1.099	0.693	0.405	0.000	−0.693



There is quite a good straight line with the equation $E = 0.0258 \times \ln(\text{ppm}) + 0.2467$. Hence A = 0.2467 V and B = 0.0258 V.
Rearranging the equation of the line gives

$$\ln(\text{ppm}) = \frac{E - 0.2467}{0.0258}$$

hence

$$\text{ppm} = \exp\left(\frac{E - 0.2467}{0.0258}\right)$$

Using these expressions the quoted potentials can be converted to ppm

E / V	0.254	0.255	0.256
ln (ppm)	0.282	0.321	0.360
ppm	1.326	1.379	1.433

The average F^- content is 1.38 ppm, with the measurements spread by about ±0.05 ppm about this average.

21.11

In a glass electrode, the potential across the thin glass membrane results from some of the Na^+ ions in the glass exchanging with H^+ ions in the solution. Like any equilibrium, this can be perturbed by adding an excess of one of the 'reagents' or 'products'. Added Na^+ ions in solution presumably compete with the H^+ ions to exchange with the ions in the glass, thereby altering the potential from what it would have been in the absence of extra Na^+ ions in solution.

21.12

(a) RHS – LHS gives the conventional reaction as

$$\mathrm{Cl^-(aq)_L} \longrightarrow \mathrm{Cl^-(aq)_R}.$$

The Nernst equation is therefore ($n = 1$)

$$\begin{aligned} E &= -\frac{RT}{F}\ln\frac{\left([\mathrm{Cl^-_R}]/c^\circ\right)}{\left([\mathrm{Cl^-_L}]/c^\circ\right)} \\ &= \frac{RT}{F}\ln\frac{[\mathrm{Cl^-_L}]}{[\mathrm{Cl^-_R}]} \end{aligned}$$

The standard half cell potentials cancel out as they are identical; to go to the second line we have used the identity $-\ln(x) \equiv \ln(1/x)$, and we have also cancelled the c° terms.

(b) The standard half cell potential of AgCl/Cl^- does not appear as the RHS and LHS half cells are the same, so $E^\circ_R - E^\circ_L = 0$.

(c) If $[\mathrm{Cl^-(aq)_R}] > [\mathrm{Cl^-(aq)_L}]$ the ratio $[\mathrm{Cl^-(aq)_L}]/[\mathrm{Cl^-(aq)_R}]$ is less than one, so the ln term in the Nernst equation, and hence the cell potential, is negative. This means that the spontaneous reaction is the *reverse* of the conventional reaction

$$[\mathrm{Cl^-(aq)_R}] > [\mathrm{Cl^-(aq)_L}] \quad \text{spontaneous} \quad \mathrm{Cl^-(aq)_R} \longrightarrow \mathrm{Cl^-(aq)_L}.$$

This reaction means that Cl^- is taken from the RHS to the LHS i.e. from the more concentrated solution to the less concentrated one. Such a process will eventually equalize the concentrations of the solutions, at which point $E = 0$ and no further change takes place. In other words, the spontaneous cell reaction is taking us to equilibrium.

If $[\mathrm{Cl^-(aq)_L}] > [\mathrm{Cl^-(aq)_R}]$ the ratio $[\mathrm{Cl^-(aq)_L}]/[\mathrm{Cl^-(aq)_R}]$ is greater than one, so the ln term in the Nernst equation, and hence the cell potential, is positive. This means that the spontaneous reaction is the *same as* of the conventional reaction

$$[\mathrm{Cl^-(aq)_L}] > [\mathrm{Cl^-(aq)_R}] \quad \text{spontaneous} \quad \mathrm{Cl^-(aq)_L} \longrightarrow \mathrm{Cl^-(aq)_R}.$$

Once again, the Cl^- is going from the more concentrated solution to the less concentrated one, taking us towards equilibrium.

(d)

$$\begin{aligned}\frac{dE}{dT} &= \frac{d}{dT}\frac{RT}{F}\ln\frac{[Cl^-_L]}{[Cl^-_R]}\\ &= \frac{R}{F}\ln\frac{[Cl^-_L]}{[Cl^-_R]}.\end{aligned}$$

To go to the second line we have recognised that all the terms on the right apart from T do not depend on temperature. The function we have to take the derivative of is just (const. $\times\, T$).

We compute $\Delta_r S_{cell}$ in the usual way (section 21.2.3 on page 798), noting that $n = 1$:

$$\begin{aligned}\Delta_r S_{cell} &= nF\frac{dE}{dT}\\ &= 1\times F\times\frac{R}{F}\ln\frac{[Cl^-_L]}{[Cl^-_R]}\\ &= R\ln\frac{[Cl^-_L]}{[Cl^-_R]}\end{aligned}$$

From the above expression for E we obtain

$$\begin{aligned}\Delta_r G_{cell} &= -nFE\\ &= -1\times F\times\frac{RT}{F}\ln\frac{[Cl^-_L]}{[Cl^-_R]}\\ &= -RT\ln\frac{[Cl^-_L]}{[Cl^-_R]}\end{aligned}$$

Given that $\Delta_r G = \Delta_r H - T\Delta_r S$ we have

$$\begin{aligned}\Delta_r H_{cell} &= \Delta_r G_{cell} + T\Delta_r S_{cell}\\ &= -RT\ln\frac{[Cl^-_L]}{[Cl^-_R]} + T\times R\ln\frac{[Cl^-_L]}{[Cl^-_R]}\\ &= 0.\end{aligned}$$

Since we have assumed that the solutions are ideal, simple decreasing the concentration of one at the expense of the other is not expected to be accompanied by any change in energy, hence $\Delta_r H_{cell} = 0$. However, there is an increase in entropy associated with equalizing the concentrations of the two solutions (just in the same way that the mixing of two gases is accompanied by an increase in entropy). In other words, the cell can be regarded as being 'driven' by the increase in entropy of the system, with there being no entropy change of the surroundings.

21.13

(a)

$$\text{RHS}\quad Hg_2Cl_2(s) + 2e^- \longrightarrow 2Hg(l) + 2Cl^-(aq)$$

$$\text{LHS}\quad AgCl(s) + e^- \longrightarrow Ag(m) + Cl^-(aq)$$

Conventional = RHS − 2 × LHS

$$Hg_2Cl_2(s) + 2Ag(m) \longrightarrow 2Hg(l) + 2AgCl(s)$$

Note that, as the same $Cl^-(aq)$ solution is shared by both electrodes, the species Cl^- cancels out when we compute the conventional cell reaction. The Nernst equation is:

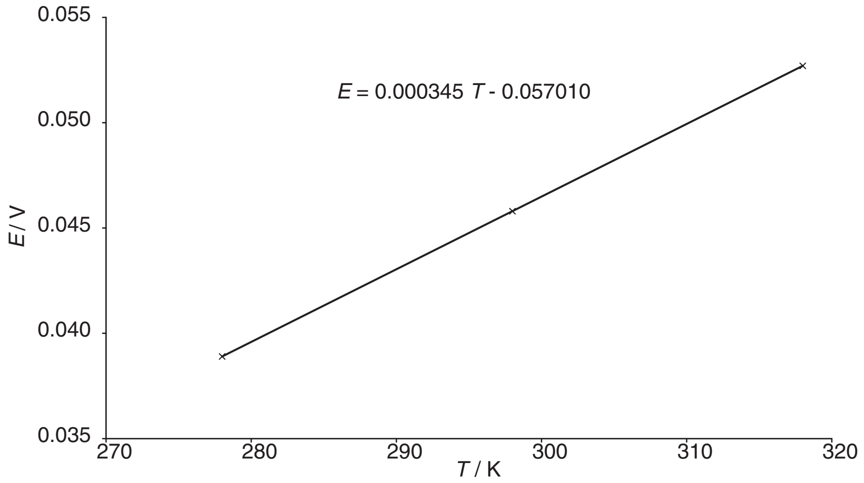
$$E = E^\circ(Hg_2Cl_2/Cl^-) - E^\circ(AgCl/Cl^-).$$

As already noted, no concentration term for Cl^- appears since this species is common to the RHS and LHS, and so does not appear in the cell reaction. All of the other species are solids or pure liquids and so do not contribute concentration terms.

(b) Since for this cell $E = E^\circ_{cell}$, from the data given E° at 298 K is 0.0458 V, and hence, recalling that $n = 2$,

$$\Delta_r G^\circ_{cell} = -nFE^\circ = -1 \times 96485 \times 0.0458 = -4419 \text{ J mol}^{-1}.$$

(c) We draw a graph of E (which in this case is E°) *vs* T



The slope is dE°/dT, and is 3.45×10^{-4} V K^{-1}. Using the results in section 21.2.3 on page 798 we have

$$\begin{aligned}\Delta_r S^\circ_{cell} &= nF\frac{dE^\circ}{dT} \\ &= 2 \times 96485 \times \left(3.45 \times 10^{-4}\right) \\ &= 66.6 \text{ J K}^{-1} \text{ mol}^{-1}.\end{aligned}$$

Given that $\Delta_r G^\circ_{cell} = \Delta_r H^\circ_{cell} - T\Delta_r S^\circ_{cell}$ we have

$$\begin{aligned}\Delta_r H^\circ_{cell} &= \Delta_r G^\circ_{cell} + T\Delta_r S^\circ_{cell} \\ &= -4419 + 298 \times 66.6 \\ &= 1.54 \times 10^4 \text{ J mol}^{-1} \\ &= 15.4 \text{ kJ mol}^{-1}.\end{aligned}$$

(d) From the cell reaction we have

$$\Delta_r H^\circ_{cell} = 2\underbrace{\Delta_f H^\circ(Hg(l))}_{=0} + 2\Delta_f H^\circ(AgCl(s)) - \Delta_f H^\circ(Hg_2Cl_2(s)) - 2\underbrace{\Delta_f H^\circ(Ag(m))}_{=0}$$

The indicated standard enthalpies of formation are zero as they are for elements in their reference phases. The required value for $\Delta_f H^\circ(Hg_2Cl_2(s))$ is therefore

$$\begin{aligned}\Delta_f H^\circ(Hg_2Cl_2(s)) &= 2\Delta_f H^\circ(AgCl(s)) - \Delta_r H^\circ_{cell} \\ &= 2 \times (-126.8) - (15.4) \\ &= -269 \text{ kJ mol}^{-1}.\end{aligned}$$

21.14

(a)

$$\text{RHS} \quad PbSO_4(s) + 2e^- \longrightarrow Pb(s) + SO_4^{2-}(aq)$$

$$\text{LHS} \quad H^+(aq) + e^- \longrightarrow \tfrac{1}{2}H_2(g)$$

Conventional = RHS − 2 × LHS

$$PbSO_4(s) + H_2(g) \longrightarrow Pb(s) + SO_4^{2-}(aq) + 2H^+(aq)$$

(b) We know the standard EMF of the following half cell, as it is the standard EMF of the cell given at the head of the question since in that cell the LHS is the SHE.

$$PbSO_4(s) + 2e^- \longrightarrow Pb(s) + SO_4^{2-}(aq) \quad E^\circ(PbSO_4/SO_4^{2-}) = -0.356\ \text{V}.$$

We are also given the standard potential for the Pb^{2+}/Pb electrode, which refers to the half cell

$$Pb^{2+}(aq) + 2e^- \longrightarrow Pb(s) \quad E^\circ(Pb^{2+}/Pb) = -0.126\ \text{V}.$$

If we imagine a cell in which the RHS is the $PbSO_4/SO_4^{2-}$ electrode and the LHS is the Pb^{2+}/Pb electrode, then the conventional cell reaction of this cell will be

$$PbSO_4(s) \longrightarrow Pb^{2+}(aq) + SO_4^{2-}(aq) \qquad (21.\text{VI})$$

The corresponding standard cell potential will be

$$E^\circ_{\text{cell}} = E^\circ(PbSO_4/SO_4^{2-}) - E^\circ(Pb^{2+}/Pb) = -0.356 - (-.126) = -0.230\ \text{V},$$

and the value of $\Delta_r G^\circ_{\text{cell}}$ is

$$\begin{aligned} \Delta_r G^\circ_{\text{cell}} &= -nFE^\circ \\ &= -2 \times 96485 \times (-0.230) \\ &= 4.44 \times 10^4\ \text{J mol}^{-1} \\ &= 44.4\ \text{kJ mol}^{-1}. \end{aligned}$$

The solubility product of $PbSO_4$ *is* the equilibrium constant for the reaction of Eq. 21.VI, for which $K = \left([Pb^{2+}]/c^\circ\right)\left([SO_4^{2-}]/c^\circ\right)$. This equilibrium constant can be computed from the value of $\Delta_r G^\circ_{\text{cell}}$ just determined. Given that $\Delta_r G^\circ = -RT \ln K$ it follows that

$$\begin{aligned} K &= \exp\left(\frac{-\Delta_r G^\circ_{\text{cell}}}{RT}\right) \\ &= \exp\left(\frac{-44.4 \times 10^3}{8.3145 \times 298}\right) \\ &= 1.65 \times 10^{-8}. \end{aligned}$$

If all that is happening is that $PbSO_4$ is dissolving to Pb^{2+} and SO_4^{2-}, it follows that $[Pb^{2+}] = [SO_4^{2-}]$. We therefore have

$$K = ([Pb^{2+}]/c^\circ)^2.$$

It follows that $[Pb^{2+}] = c^\circ \times \sqrt{K}$, which is 1.28×10^{-4} mol dm^{-3} (recall that $c^\circ = 1$ mol dm^{-3}).

(c) Since $dE^\circ/dT = 1.1 \times 10^{-4}$ V K^{-1}, we can compute $\Delta_r S^\circ_{cell}$ in the usual way (section 21.2.3 on page 798)

$$\begin{aligned}\Delta_r S^\circ_{cell} &= nF\frac{dE^\circ}{dT} \\ &= 2 \times 96485 \times \left(1.1 \times 10^{-4}\right) \\ &= 21.2 \text{ J K}^{-1} \text{ mol}^{-1}.\end{aligned}$$

From the given value of the standard cell potential of −0.356 V we can find the standard Gibbs energy change of this reaction as (noting that $n = 2$)

$$\begin{aligned}\Delta_r G^\circ_{cell} &= -nFE^\circ \\ &= -2 \times 96485 \times (-0.356) \\ &= 6.87 \times 10^4 \text{ J mol}^{-1} \\ &= 68.7 \text{ kJ mol}^{-1}.\end{aligned}$$

Given that $\Delta_r G^\circ_{cell} = \Delta_r H^\circ_{cell} - T\Delta_r S^\circ_{cell}$ we have

$$\begin{aligned}\Delta_r H^\circ_{cell} &= \Delta_r G^\circ_{cell} + T\Delta_r S^\circ_{cell} \\ &= 68.7 \times 10^3 + 298 \times 21.2 \\ &= 7.50 \times 10^4 \text{ J mol}^{-1} \\ &= 75.0 \text{ kJ mol}^{-1}.\end{aligned}$$

(d) Referring back to the cell reaction we have

$$PbSO_4(s) + H_2(g) \longrightarrow Pb(s) + SO_4^{2-}(aq) + 2H^+(aq)$$

for which

$$\Delta_r H^\circ_{cell} = \underbrace{\Delta_f H^\circ(Pb(s))}_{=0} + \Delta_f H^\circ(SO_4^{2-}(aq)) + 2\underbrace{\Delta_f H^\circ(H^+(aq))}_{=0} - \Delta_f H^\circ(PbSO_4(s)) - \underbrace{\Delta_f H^\circ(H_2(g))}_{=0}$$

The first and last terms are zero as they are the standard enthalpies of formation of elements in their reference phases. The second term is also zero by convention, see section 21.9.2 on page 817.

Using the data we already have we can rearrange this to find $\Delta_f H^\circ(SO_4^{2-}(aq))$

$$\begin{aligned}\Delta_f H^\circ(SO_4^{2-}(aq)) &= \Delta_r H^\circ_{cell} + \Delta_f H^\circ(PbSO_4(s)) \\ &= 75.0 + (-918.4) \\ &= -843.4 \text{ kJ mol}^{-1}.\end{aligned}$$

21.15

(a) The standard electrode potential is the potential developed by a cell when all of the species are present under standard conditions (the temperature must be stated). Standard conditions means solids and liquids in their pure forms, solutions at the standard concentration of 1 mol dm^{-3} (assuming ideality), and gases at the standard pressure of 1 bar.

The standard half-cell potential of a particular redox couple is the potential developed by a cell in which the RHS is the half-cell under consideration, with all species under standard conditions, and the LHS is the standard hydrogen electrode (SHE). The temperature must be stated.

By adopting this definition we are essentially referencing all of the half-cell potentials to that of the SHE i.e. assuming that the SHE has zero potential. This means that although we can only measure a cell potential, we can relate this measurement to a half-cell potential.

(b)

$$\mathrm{Pt(m)|H_2(g)|H^+(aq)||Ag^+(aq)|Ag(m)}$$

$$\text{RHS} - \text{LHS} \quad \mathrm{Ag^+(aq)} + \tfrac{1}{2}\mathrm{H_2(g)} \longrightarrow \mathrm{Ag(m)} + \mathrm{H^+(aq)} \qquad n = 1$$

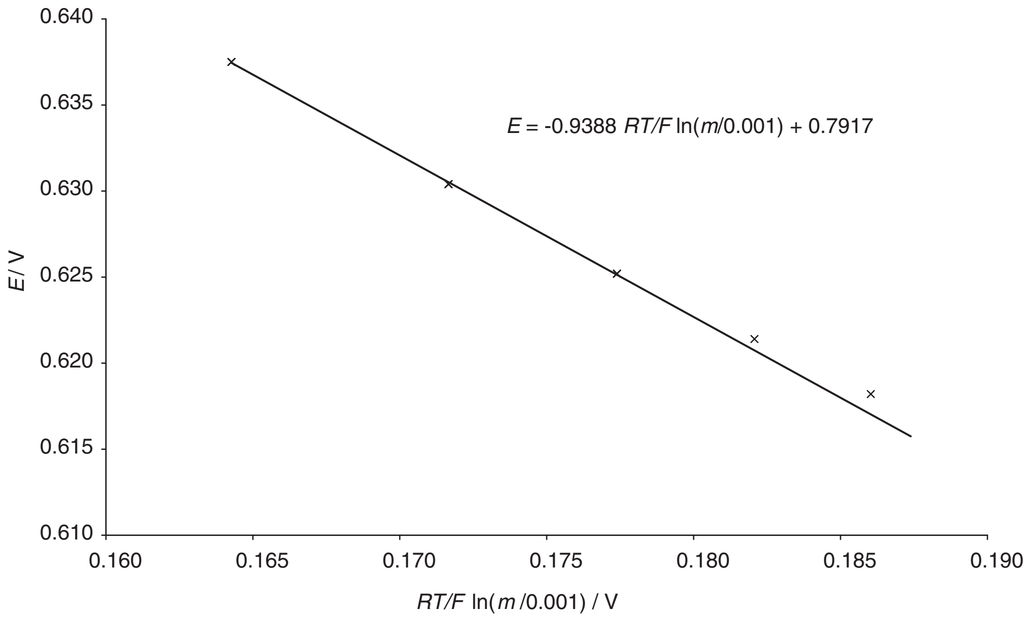
$$E = E^\circ(\mathrm{Ag^+/Ag}) - \frac{RT}{F}\ln\frac{([\mathrm{H^+}]/c^\circ)}{(p_{\mathrm{H_2}}/p^\circ)^{\frac{1}{2}}\,([\mathrm{Ag^+}]/c^\circ)}$$

Putting $p_{\mathrm{H_2}} = 1$ bar, $[\mathrm{Ag^+}] = 0.001$ mol dm^{-3}, and $[\mathrm{H^+}] = m$ mol dm^{-3} gives

$$E = E^\circ(\mathrm{Ag^+/Ag}) - \frac{RT}{F}\ln\frac{m}{0.001}$$

where the c° terms have cancelled.

This equation implies that a plot of E against $(RT/F)\ln(m/0.001)$ should be a straight line with intercept E° when $(RT/F)\ln(m/0.001) = 0$.



The graph is not a very good straight line, almost certainly due to the fact that the assumption of ideality for the solutions is not holding well, particularly at the higher concentrations.

(c) The three data points for the lower concentrations are on a reasonable straight line, and this is the line shown on the graph (we expect the assumption of ideality to be more valid for lower conentrations). The intercept of this line with the y-axis is 0.792 V, which is our estimate for $E^\circ(\mathrm{Ag^+/Ag})$

Further questions

21.16 Consider the following cell

$$\mathrm{Pt(m)\,|\,O_2(g, p_{O_2} = p_L)\,|\,NaOH(aq, 0.1\ mol\ dm^{-3})\,||\,NaOH(aq, 0.01\ mol\ dm^{-3})|O_2(g, p_{O_2} = p_R)|Pt(m)}$$

(a) Write down the conventional cell reaction and the Nernst equation for the cell.

(b) If the pressure of oxygen on both sides is 1 bar, calculate the cell potential, $\Delta_r G_{cell}$, $\Delta_r S_{cell}$ and $\Delta_r H_{cell}$.

(c) If the pressure of O_2 in the left-hand half-cell is held at 1 bar, calculate what pressure of O_2 in the right-hand half cell would be required to reduce the cell potential to zero.

21.16

(a)

$$\begin{array}{rl} \text{RHS} & O_2(g)_R + 2H_2O(l) + 4e^- \longrightarrow 4OH^-(aq)_R \\ \text{LHS} & O_2(g)_L + 2H_2O(l) + 4e^- \longrightarrow 4OH^-(aq)_L \\ \text{conventional} & O_2(g)_R + 4OH^-(aq)_L \longrightarrow 4OH^-(aq)_R + O_2(g)_L \end{array}$$

$$E = -\frac{RT}{4F} \ln \frac{\left([OH^-_R]/c^\circ\right)^4 (p_L/p^\circ)}{\left([OH^-_L]/c^\circ\right)^4 (p_R/p^\circ)}$$

(b) If $p_L = p_R$ the pressure terms cancel. We put $\left([OH^-_R]/c^\circ\right) = 0.01$ and $\left([OH^-_L]/c^\circ\right) = 0.1$ (since $c^\circ = 1 \text{ mol dm}^{-3}$), to give

$$\begin{aligned} E &= -\frac{RT}{4F} \ln \frac{\left([OH^-_R]/c^\circ\right)^4 (p_L/p^\circ)}{\left([OH^-_L]/c^\circ\right)^4 (p_R/p^\circ)} \\ &= \frac{RT}{4F} \ln \frac{(0.01)^4}{(0.1)^4} \\ &= \frac{RT}{F} \ln \frac{(0.01)}{(0.1)} \\ &= \frac{RT}{F} \ln \frac{1}{10} \\ &= -\frac{RT}{F} \ln 10. \end{aligned}$$

To go to the third line we have used $\ln(x^n) \equiv n \ln(x)$, and to go to the last line we have used $\ln(1/x) \equiv -\ln(x)$.

Recalling that $n = 4$ we have

$$\begin{aligned} \Delta_r G_{cell} &= -nFE \\ &= -4 \times F \times \left(-\frac{RT}{F} \ln 10\right) \\ &= 4RT \ln 10 \\ &= 2.28 \times 10^4 \text{ J mol}^{-1} \\ &= 22.8 \text{ kJ mol}^{-1}. \end{aligned}$$

$$\begin{aligned} \Delta_r S_{cell} &= nF \frac{dE}{dT} \\ &= 4 \times F \times \frac{d E}{dT} \left(-\frac{RT}{F} \ln 10\right) \\ &= 4F\left(-\frac{R}{F} \ln 10\right) \\ &= -4R \ln 10 \\ &= -76.5 \text{ J K}^{-1} \text{ mol}^{-1} \end{aligned}$$

$$\begin{aligned}\Delta_r H_{cell} &= \Delta_r G_{cell} + T\Delta_r S_{cell} \\ &= 4RT \ln 10 + T(-4R\ln 10) \\ &= 0\end{aligned}$$

As before, for a concentration cell we expect there to be no enthalpy change.

(c) Setting $p_L/p^\circ = 1$ and the potential to zero gives

$$\begin{aligned}0 &= -\frac{RT}{4F}\ln\frac{\left([OH_R^-]/c^\circ\right)^4}{\left([OH_L^-]/c^\circ\right)^4 (p_R/p^\circ)} \\ &= \frac{RT}{4F}\ln\frac{(0.01)^4}{(0.1)^4 (p_R/p^\circ)} \\ &= \frac{RT}{4F}\ln\frac{1}{10^4 (p_R/p^\circ)}\end{aligned}$$

For the RHS to be zero, the term in the ln must be 1 (since $\ln 1 = 0$):

$$\begin{aligned}1 &= \frac{1}{10^4 (p_R/p^\circ)} \\ \text{hence } (p_R/p^\circ) &= 10^{-4}\end{aligned}$$

It thus follows that $p_R = 10^{-4}$ bar is the pressure needed to reduce the potential to zero.